

By: AKH
 Rev.--- By: ---
 Chk: ---

Date Jun 2026
 Date ---
 Date ---

MacTest

Macaulay Beam Analysis - Example Worksheet

101 - (2) EQ Span - Point Load

Beam Sect = W14x61 W14x90
 A = 17.90 26.50 in²
 I = 640.00 999.00 in⁴

E = 29,000,000 lb / in²

Length Units in
 Force Units lb

Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E
		in	in ²	in ⁴	lb / in ²
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07
	2				
	3				
	4				
	5				

Supports

	Supp. No.	X	Type	Imposed Movement	
		in	---	* dY (vert)	** dr (rot)
DO NOT DELETE ROWS (GRAPHED)	1	0.0	P	0.0	0.000
	2	300.0	P	0.0	0.000
	3	600.0	P	0.0	0.000
	4				
	5				

* (settlement) ** (Not Working)

Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)
		in	in	lb / in	lb / in
DO NOT DELETE ROWS (GRAPHED)	1	0.0	600.0	0.0	0.0
	2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0
	5	0.0	0.0	0.0	0.0

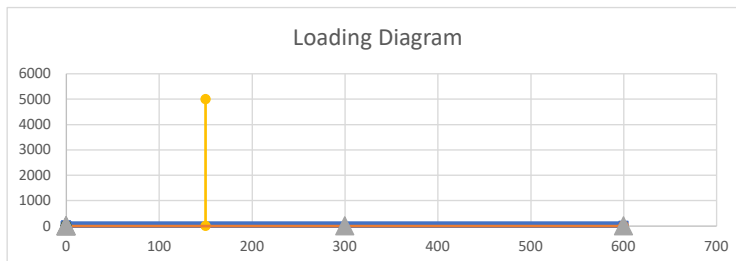
Load
lb / ft
0.0
0.0
0.0
0.0

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	150	-5000.00	0.00
	2	0	0.00	0.00
	3			
	4			
	5			

My Load
lb-ft
0.0
0.0
0.0
0.0

Loading Diagram



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BEAM REACTIONS

Position (X)	Enforced Disp (dy)	Enforced Rot (theta)	Vert Reac (Ry)	Internal Moment (Mz)	Type
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	2,031.2	0.0	P
300.00	0.00	0.00	3,437.6	0.0	P
600.00	0.00	0.00	-468.8	0.0	P

Vert Reac
(kip)
2.031
3.438

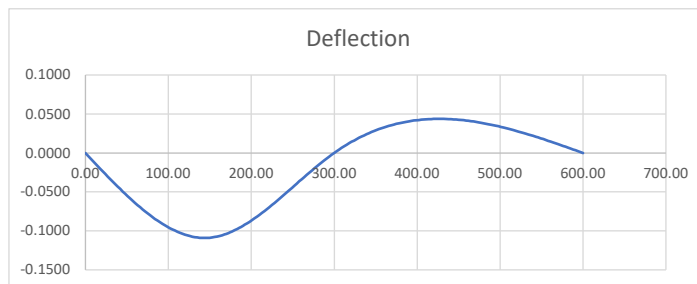
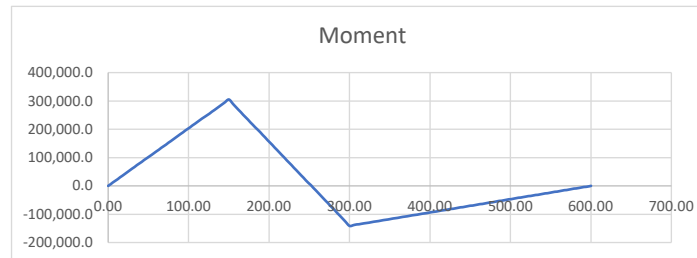
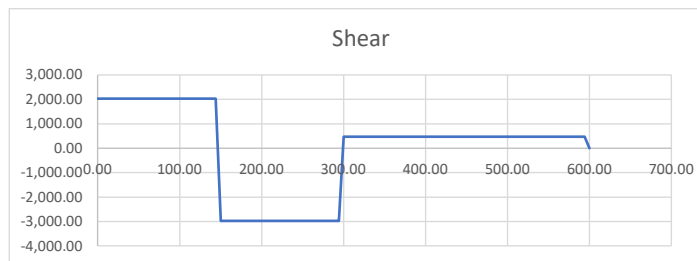
BEAM REACTIONS EQUILIBRIUM CHECK

Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-5,000.0	5,000.0	0.0	OK
Sum M @ x=0	F-L	-7.5000E+05	7.5000E+05	0.0	OK

BEAM MAX/MIN LOAD EFFECTS

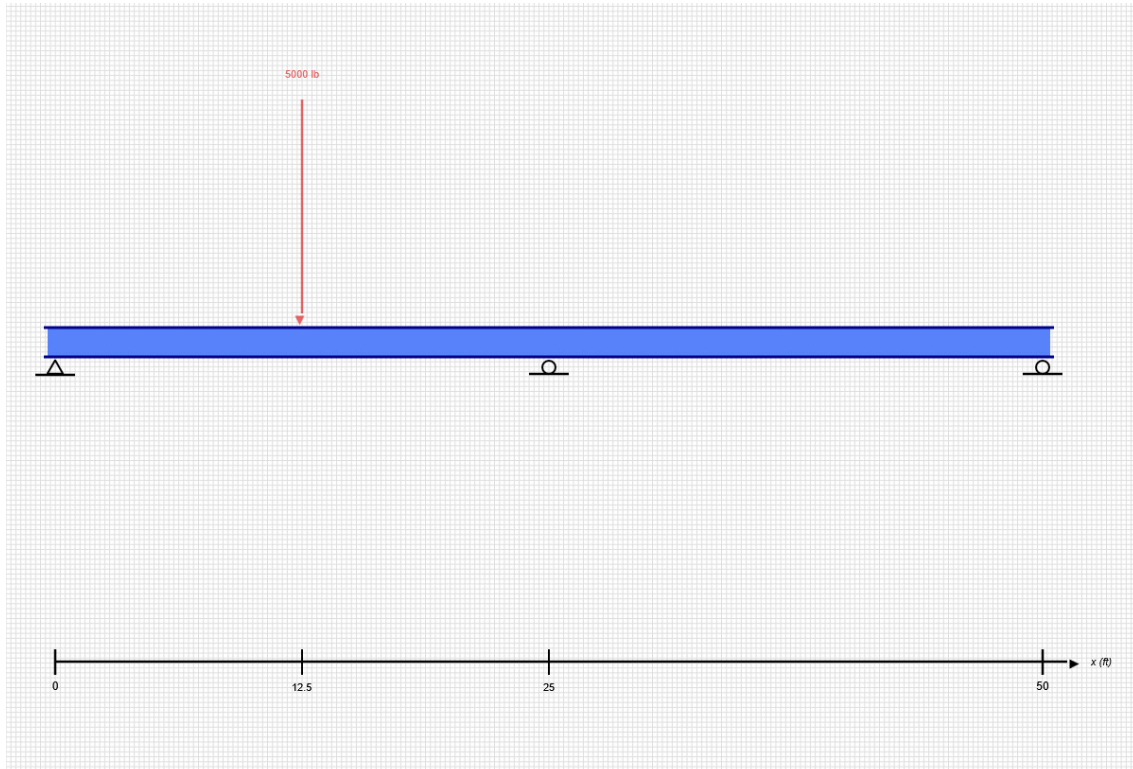
Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	0.00	2,031.2	0.0	0.0000
Max -V	150.00	-2,968.8	304,680.5	-0.1089
Max +M	150.00	-2,968.8	304,680.5	-0.1089
Max -M	300.00	468.8	-140,639.1	0.0000
Max Defl	144.00	2,031.2	292,493.2	-0.1091

Max/Min Value
(---)
2.03 kip
-2.97 kip
25.39 kip-ft
-11.72 kip-ft
-0.109 in



SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2
Thu Jun 18 2026 11:25:37 GMT-0400 (Eastern Daylight Time)

Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results
- Bending Moment Diagram (BMD)
- Shear Force Diagram (SFD)
- Deflection Results

SKYCIV FREE TRIAL EDITION

INPUT SUMMARY

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

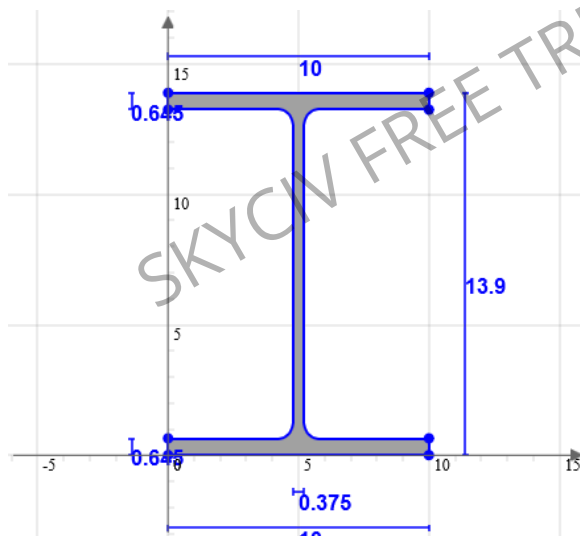
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft
Roller	25 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	12.5 ft	-5000 lb	DL

Beam Section



Geometric Properties		
A	17.9	in ²
C _z	5	in
C _y	6.95	in

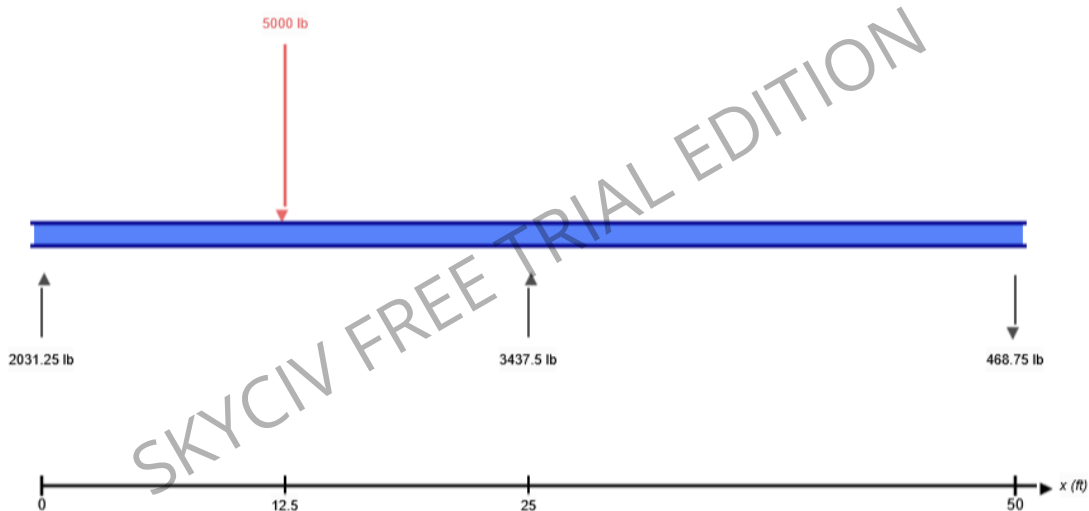
Bending Properties		
I _z	640	in ⁴
I _y	107	in ⁴

Shear Properties		
A _z	11.277	in ²
A _y	4.99	in ²

Torsion Properties		
J	2.19	in ⁴
r	0.991	in

Shape	Material	E (ksi)	ν	ρ (lb/ft ³)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.045	L/5508
Custom Stress Limit	PASS	39 ksi	0.085	3.309 ksi
Material Yield	PASS	36 ksi	0.092	3.309 ksi
Material Strength	PASS	58 ksi	0.057	3.309 ksi

ANALYSIS RESULTS

Reactions

Support at	R _x (lb)	R _y (lb)	M (kip-ft)
0	0	2031.25	0
25	0	3437.5	0
50	0	-468.75	0

Force Extremes

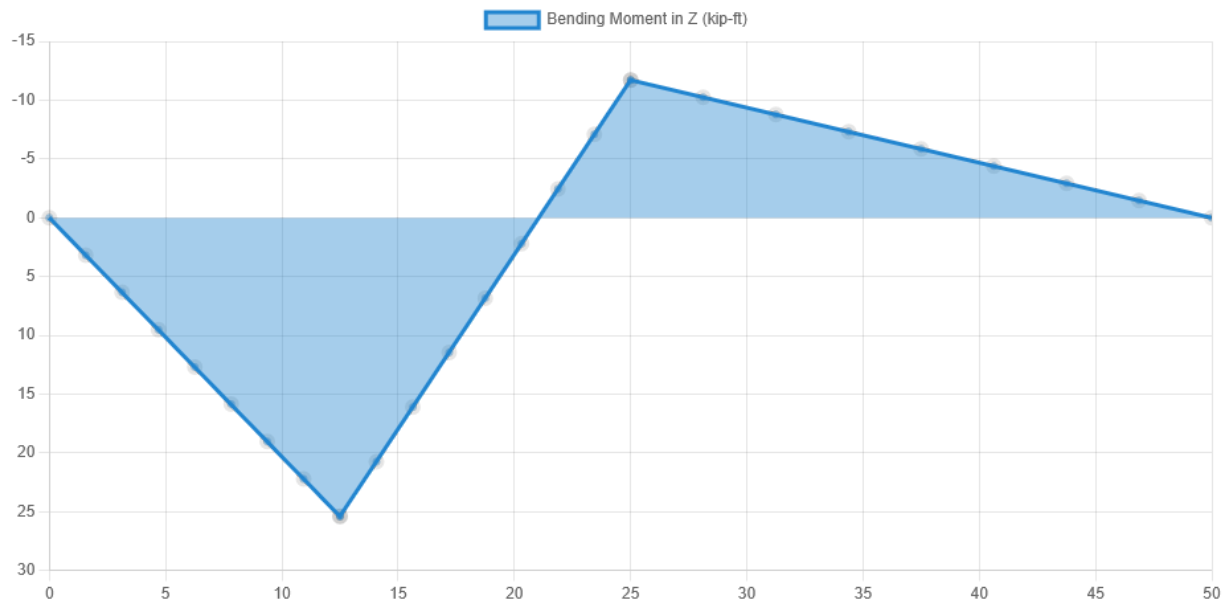
Result	Max	Min
Bending Moment	25.391 kip-ft	-11.719 kip-ft
Shear	2031.25 lb	-2968.75 lb
Displacement Y	0.043 in	-0.109 in

Stress Extremes

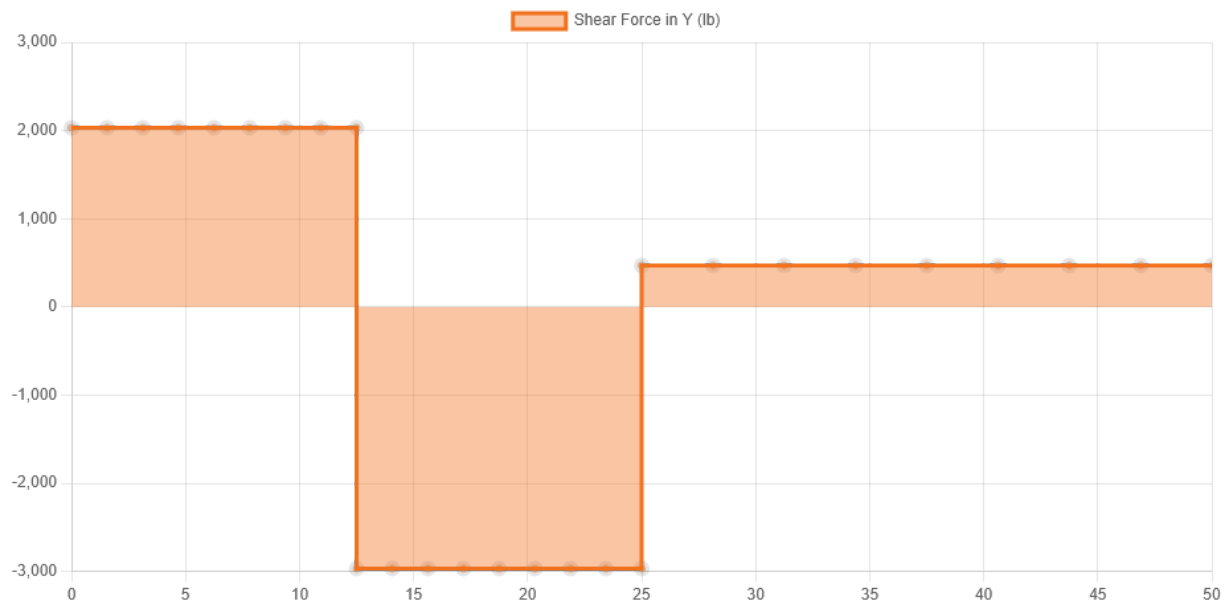
Result	Max (ksi)	Min (ksi)
Bending Stress	3.309	-3.309
Shear Stress Total	0.634	0.1
Max Combined Normal Stress	3.309	0
Min Combined Normal Stress	0	-3.309

DIAGRAMS

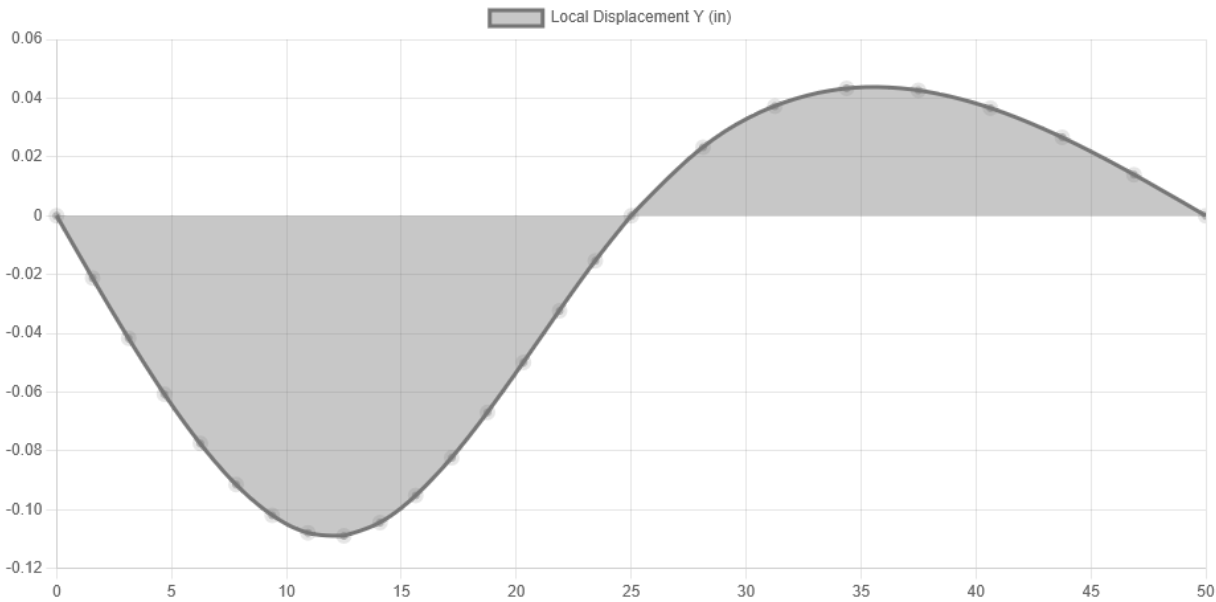
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	0	12.5	25	34.375	50
Total Deflection (in)	0 ?	0.109 ?	0 ?	0.043 ?	0 ?
Deflection/Span	-	L/2754	-	L/6930	-
The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings.					